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The brief Experiences in Close Relationships Scale - Revised Child version (ECR-RC): Factor structure and invariance across middle childhood and early adolescence

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Abstract

The recently developed short form of the Experiences in Close Relationships Scale-Revised Child version (ECR-RC) is a promising tool to assess anxious and avoidant attachment in children and adolescents. Yet, evidence concerning its validity in middle childhood is limited. This study aimed to test the psychometric properties of the 12-item ECR-RC for both mother and father forms in a sample of 448 Italian children (50.2% girls) aged between 8 and 13 years. The scale was adapted by changing the response format to make it more understandable for young children. Psychometric proprieties of the brief ECR-RC were investigated by testing its factor structure and internal consistency, invariance across middle childhood and early adolescence, and concurrent and convergent validity. A series of confirmatory factor analyses provided support for the two-factor structure (i.e., anxiety and avoidance) of the ECR-RC, and multi-group confirmatory factor analyses supported its invariance across middle childhood and early adolescence. Older children reported significantly higher latent mean values in avoidant attachment to both parents compared to their younger counterparts. Furthermore, the questionnaire showed evidence of concurrent and convergent validity. Our results indicate that the 12-item version of the ECR-RC is a psychometrically robust instrument to assess avoidance and anxiety toward mother and father among Italian children and early adolescents.

Keywords

ECR-RC, psychometric properties, attachment, middle childhood, early adolescence

Introduction

Insecure attachment has been recognized as a risk factor for socio-emotional development across the life span. Indeed, empirical evidence suggests that insecure attachment patterns in infancy predict difficulties in emotion regulation (ER) and problem behaviors later in childhood (Groh, Roisman, van Ijzendoorn, Bakermans-Kranenburg, & Fearon, 2012). Similar findings have been reported for adolescent and adult samples, in which avoidance and anxiety were found to be differentially related to a wide variety of mental and/or personality disorders (Mikulincer & Shaver, 2012). However, less is known about how different patterns of insecurity may influence developmental outcomes in middle childhood, partly because there are few well-validated measures for distinguishing anxiety and avoidance dimensions during this understudied developmental stage (Brenning, Soenens, Braet, & Bosmans, 2011).

Middle childhood (ranging approximately from age 6 to 10) and early adolescence (ranging approximately from age 11 to 14) represent critical developmental periods (Eccles, 1999), because children show a substantial increase in abstract thinking, cognitive flexibility, and metacognitive skills, as well as a progressive improvement in ER skills (Raikes & Thompson, 2005). Furthermore, they become more independent, self-aware, and involved in social contexts beyond the family, and begin to conceive of relationships in more complex forms (Raikes & Thompson, 2005). Thus, their "attachment representations" become more elaborate and organized (Kerns & Brumariu, 2016), rendering the separation and reunion procedures used in early childhood (e.g., Strange Situation; Ainsworth, Blehar, Waters, & Wall, 1978) less suitable for use with children at these ages.

Over the past 20 years, several instruments have been developed to assess attachment in middle childhood. Most of these are "representational" attachment measures (Kerns & Brumariu, 2016), such as interviews (e.g., Child Attachment Interview; Shmueli-Goetz, Target, Fonagy, & Datta, 2008), narrative story stem techniques (e.g., Attachment Doll Story Completion Task; Granot, & Mayseless, 2001), and self-report questionnaires (e.g., Security Scale (SS); Kerns, Aspelmeier, Gentzler, & Grabill, 2001). Although these measures have provided a substantial contribution to the study of attachment, "there is currently no dominant conceptual or methodological approach" (Kerns & Brumariu, 2016, p. 351) regarding the conceptualization and measurement of attachment in middle childhood. Despite the challenges inherent in the use of subjective attachment measures, Bosmans and Kerns (2015)

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recently advocated the usefulness of this measurement approach in middle childhood. Referring to dual process theory (e.g., Gawronski & Creighton, 2013), they argued that available measures capture specific aspects of the attachment system (Bosmans & Kerns, 2015; Psouni & Apetroaia, 2014) because they tap into different processes, namely strategic (i.e., explicit and/or available for conscious introspection) and automatic (i.e., outside conscious control) processes. As a consequence, they should be conceived of as complementary rather than overlapping and may not necessarily correlate (Moors & de Houwer, 2006). The cognitive and emotional achievements occurring in middle childhood result in a more accurate understanding and management of children's feelings, internal states, and manifest behaviors. In this perspective, self-report questionnaires can be considered a valid approach to investigate attachment organization as well as attachment-related outcomes. Thus, the use of different measures based on different approaches is recommended to obtain a more detailed understanding about the meaning of attachment in this in-between developmental stage (Bosmans & Kerns, 2015).

Among existing self-report measures of attachment, only a few were specifically designed for use in middle childhood (i.e., SS; Kerns et al., 2001 and Coping Strategies Questionnaire; Finnegan, Hodges, & Perry, 1996). Others were originally developed to measure attachment in older adolescents (e.g., Inventory of Parent and Peer Attachment; Armsden & Greenberg, 1987) and adults (e.g., Experiences in Close Relationships (ECR) Scale; Fraley, Waller, & Brennan, 2000), and have been simplified and adapted for use in the context of parent-child relationships. In this paper, we focus on the Experiences in Close Relationships-Revised Child version (ECR-RC; Brenning, van Petegem, Vanhalst, & Soenens, 2014). The scale was recently adapted from the widely validated ECR, and measures attachment anxiety and avoidance in children and adolescents. A long and a short version of the ECR-RC are currently available.

The long version (Brenning, Soenens, Braet, & Bosmans, 2011) includes all 36 items of the original ECR designed for adults. In accordance with the adult version, each question is rated on a seven-point Likert-type scale ranging from 1 (''strongly disagree'') to 7 (''strongly agree''). Exploratory and confirmatory factor analyses (CFAs) among children aged 8–13 years provided support for a two-factor structure, representing attachment anxiety and avoidance. Invariance was established across gender and age groups (8–10 vs. 11–13 years), and both avoidance and anxiety subscales of the mother and father forms showed good internal consistency (Brenning et al., 2011). In addition, significant and positive correlations between the anxious and avoidant subscales and other attachment representational measures (i.e., SS, Kerns et al., 2001; Relationship Questionnaire for Children, Bartholomew & Horowitz, 1991) were found.

More recently, a short version has also been proposed. The brief ECR-RC (Brenning, van Petegem, Vanhalst, & Soenens, 2014) includes 12 items (the same for mother and father) selected from the original version, in which six items refer to anxiety, and six items measure the avoidance dimension. Each question is rated on a seven-point Likert-type scale ranging from 1 (''strongly disagree'') to 7 (''strongly agree''). Concerning its factor structure, a principal component analysis provided evidence for a two-dimensional solution (Brenning et al., 2014). CFA was performed in normative and clinical samples, yielding different solutions (Brenning et al., 2015; Lionetti, Mastrotheodoros, & Palladino, 2017). Of concern, Brenning et al. (2014) performed three different CFAs in two normative samples (N = 310, age range 13–20 years; N = 110, age range 15–18 years) and in one clinical sample (N = 99, age range 10–18 years). Results showed a reasonably good fit, corroborating the two-dimensional structure of the 12-item scale. A more recent study among 961 Italian adolescents aged between 12 and 19 years (Lionetti et al., 2017) proposed a third factor labeled "Security," which included three of the six items originally pertaining to the avoidance subscale (i.e., "I usually talk to my father/mother about my problems and worries"; "When I feel bad, it helps to talk to my father/mother"; "I tell my father/mother nearly everything"). A three-factor model solution yielded better fit indices compared to the two-factor model. Internal consistency as measured through Cronbach's Alpha coefficient reported good values for both the two- and three-factor solutions (Brenning et al., 2015; Lionetti et al., 2017). No information on test-retest reliability is currently available for the 12-item version.

With regard to convergent validity, Brenning et al. (2014) found that both attachment-related dimensions (anxiety and avoidance) were positively related to children's perceptions of parental psychological control, and negatively associated with child reports of parental responsiveness. Furthermore, meaningful associations with ER strategies emerged, with the anxiety dimension being positively related to dysregulation, and the avoidant subscale being positively correlated with suppression (Brenning, Soenens, Braet, & Bosmans, 2012). The authors explained this finding with reference to the model of Shaver and Mikulincer (2002), which posits that avoidant attachment is characterized by the endorsement of deactivating strategies that lead to the suppression of negative emotion, whereas anxious attachment mostly involves the use of hyper-activating ER strategies. Conversely, secure attachment is marked by a more flexible thought about emotion-eliciting events and a reframing of the situation in a reality-based manner (Mikulincer & Shaver, 2007).

To summarize, the 12-item ECR-RC is a promising tool to assess attachment anxiety and avoidance in children and adolescents. However, empirical evidence concerning its factor structure is still scarce. Also, to our knowledge, only one study has explored its factor structure using CFA in children aged 8–16 years (Brenning et al., 2014), but invariance across middle childhood and early adolescence was not tested. This study aimed to address these gaps by assessing the psychometric properties of the short form of the ECR-RC for use with Italian children aged between 8 and 13 years. Specifically, we (1) evaluated the factor structure and internal consistency of the Italian version via CFA; (2) tested its invariance across middle childhood and early adolescence; and, in case scalar invariance was established, (3) analyzed group differences in anxiety and avoidance scores and (4) examined concurrent validity of the scale with a measure of perceived attachment security, and convergent validity by exploring its associations with self-worth and two ER strategies, namely cognitive reappraisal (CR) and expressive suppression (ES). Specifically, based on extant theory (see Cassidy, 1990), we expected the avoidance and anxious attachment scores to be negatively related to children's self-worth. With regard to ER strategies, because CR is an antecedent-focused strategy that attempts to reshape an emotionally eliciting event (Gross & John, 2003), we anticipated that this strategy would be negatively linked to both anxious and avoidant subscale scores. In contrast, because ES is deemed a

response-focused strategy involving the inhibition of observable expression triggered by the emotional experience (Gross & John, 2003), it was reasonable to expect that this ER strategy would show a positive association with avoidant attachment.

Furthermore, because there is evidence that seven-point Likert scales can be problematic for children (e.g., Borgers, Hox, & Sikkel, 2004), and given that a better quality in responses has been found when every point—rather than only the two extremes—of a response scale is labeled (Borgers, Hox, & Sikkel, 2003), we changed the response format of the brief ECR-RC accordingly to make it more understandable for children.

Materials and Method

Participants

The study was introduced to 520 children aged between 8 and 13 years recruited from five primary schools and three middle schools in Northeastern Italy. All children whose parents signed the informed consent form ($N = 492$, 94.6%) were involved in the study.

Twenty-six children were absent on the day of data collection. Moreover, questionnaires completed by children with intellectual disabilities or certified developmental/learning disorders (3.4%, $N = 18$) were not included in data analyses.

Hence, the final sample consisted of 448 participants, of whom 259 were primary school children (hereafter labeled "younger children"; 51% girls, mean age = 9 years and 2 months, $SD = 7$ months, range = 8.2–10.3) and 189 were middle school children (hereafter labeled "older children"; 50%, mean age = 11 years and 8 months, $SD = 4$ months, range = 11.2–12.9). The two groups of children did not differ in terms of gender distribution ($x^2(1,448) = 0.11$, $p = 0.744$, Cramer's $phi = 0.016$) or socio-economic status ($t(444) = 0.45$, $p = 0.650$, $d_{\text{Cohen}} = 0.044$) as assessed via the Family Affluence Scale (FAS; Currie, Molcho, Boyce, Holstein, Torsheim, & Richter, 2008; see next section).

In the final sample, 13 children did not fill out the father items, whereas four children did not complete the mother items (see Table A1 and Table A2 in Appendix I for more details concerning missing data).

Measures

Children's socio-economic status was assessed via the FAS (Currie et al., 2008), a widely used four-item measure of family wealth (e.g., "Does your family own a car?"; "How many times did you travel away on holiday with your family during the past 12 months?"). Scores across items are summed to provide an overall score ranging from 0 to 9, in which scores from 0 to 2 indicate low affluence, 3–5 indicate medium affluence, and 6–9 indicate high affluence. The FAS has reported good validity and reliability across different cultures and countries, including the Italian context (Vieno, Santinello, Lenzi, Baldassari, & Mirandola, 2009). In this study, the majority of participants were from middle (medium affluence: 24.5%) or upper-income families (high affluence: 71.1%).

The ECR-RC (Brenning et al., 2015) was designed to measure attachment anxiety (e.g., "I worry that my mother/father does not really love me") and attachment avoidance (e.g., "I prefer not to tell my mother/father how I feel deep down") in children and adolescents. The recently developed short form consists of 12 items (six for anxiety, six for avoidance) originally rated on a seven-point scale. Scores across items are averaged to provide an anxiety and an avoidance score, respectively, with higher scores indicating a more anxious or avoidant attachment. In this study, we used the Italian version of the questionnaire (Lionetti et al., 2017). As mentioned earlier, the response scale of the brief ECR-RC was reduced to a five-point Likert-scale (from 1 = strongly disagree to 5 = completely agree).

The SS (Kerns et al., 2001) is a self-report measure designed to assess "felt-security" in school age children. It consists of 15 items (the same for mother and father) rated using Harter's (1982) format, namely "Some kids . . . other kids . . . " After choosing the children that best fit them, participants are asked to indicate whether it is "really true" or "sort of true" for them. The two statements are presented as one item, and each item is scored from 1 to 4. Scores across items are averaged to provide a security score on a continuous dimension of security, with higher scores indicating higher levels of perceived attachment security. Previous research reported good psychometric proprieties of the Italian version of the SS, including structural invariance across parents (Marci, Lionetti, Moscardino, Pastore, & Altoé, 2017). For this study, we administered nine of the 12 items included in the Italian version (Marci et al., 2017), namely those with higher loadings. In this sample, CFAs of the nine items showed a good fit to the data (see preliminary analyses).

The Self-Perception Profile for Children (SPPC, Harter, 1985, 2012) is a 36-item questionnaire designed to assess children's perceptions of self-competence and adequacy. For our purpose, children completed the Global Self-Worth subscale. It consists of six items rated using the already mentioned Harter (1982) format. Participants are asked to choose the children who best fit them between the two presented, and then indicate whether the description is "really true" or "sort of true" for them. Each item is scored on a four-point scale ranging from 1 to 4. Scores across items are averaged to provide a global self-worth score. The questionnaire reported good psychometric properties in terms of factor structure, internal consistency, and convergent validity (Harter, 2012). In this sample, we used the Italian version of the scale (Pedrabissi, Santinello, & Scarpazza, 1988).

The Emotion Regulation Questionnaire for Children and Adolescents (ERQ-CA; Gullone & Taiffe, 2012) is a 10-item revised version of the Emotion Regulation Questionnaire (ERQ; Gross & John, 2003). It was designed to assess the use of two ER strategies, namely CR (six items, e.g., "When I want to feel happier, I think about something different") and ES (four items, e.g., "I keep my feelings to myself") in children and adolescents. Each item is rated on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Scores across items are averaged to provide two different scores (one for each subscale). Higher scores indicate greater use of the corresponding ER strategy. The ERQ-RC reported good internal consistencies, with alpha coefficients ranging from 0.82 to 0.89 for CR and from 0.69 to 0.79 for ES (Gullone & Taffe, 2012).

Procedure

Ethical approval was obtained from the Ethical Committee of the University of Padova (protocol #1838-2016). After obtaining approval from the school principals, a letter was sent to children's parents in order to explain the nature of the study. Written consent

was obtained from both parents, and verbal assent was obtained from each child before data collection. Children completed questionnaires in the classroom and in a single session during school hours under the supervision of a research assistant and in the presence of their teacher.

Data were collected anonymously to make children feel relaxed and limit social desirability effects. They were explained that they were free to interrupt their participation at any time without any consequences, and that their participation would remain confidential. The items were read aloud by the researcher to ensure comprehension and minimize the possible effect of differences in reading ability. All participants were given a certificate and thanked for taking part in the study.

Data Analyses

All analyses were performed in R (R Development Core Team, 2017).

In preliminary analyses, we used CFA on the nine items extracted from the SS to evaluate the internal structure of the scale.

For descriptive purposes, we calculated item response distributions of both mother and father forms of the ECR-RC and computed the main descriptive statistics (mean, SD, skewness) separately for younger and older children.

To test the factor structure and invariance of the ECR-RC between age groups, analyses were conducted following four steps. First, CFAs were performed to evaluate the factor structure of the mother items of the 12-item ECR-RC separately for each age group. Then, the most plausible model was tested on the father items, and internal consistency was evaluated for both forms. Second, Multi-Group Confirmatory Factor Analysis (MG-CFA) was used to test measurement invariance of the questionnaire across middle childhood and early adolescence. Last, correlations were computed to test convergent and concurrent validity of the scale.

To establish the factor structure of the brief ECR-RC, in each age group (i.e., younger and older children) we separately tested (a) a single-factor model, in which items assessing anxiety and avoidance loaded on a single latent factor to exclude the monodimensionality of the scale; then, we tested (b) the originally proposed two-factor solution, in which six items loaded on the anxiety dimension, and six items loaded on the avoidance dimension (Brenning et al., 2014); finally, we evaluated (c) a three-factor solution, with three items loading on a security factor, three items on the avoidance factor, and six items on the anxiety factor following previous research (Lionetti et al., 2017). The parameters were estimated using the Robust Diagonally Weighted Least Squares Mean and Variance (WLSMV) estimator (Rhemtulla, Brosseau-Liard, & Savalei, 2012). Contrary to the maximum likelihood (ML) or robust ML estimators, this method is specifically designed for ordinal data and provides the best option for modeling categorical or ordered data or when items are not normally distributed (Brown, 2006). Missing data (< 1%) were handled with the pairwise ML estimation method, which was developed for factor analytic models with ordinal data (Myrsini, Moustakib, Yang-Wallentina, & Jöreskog, 2012) and is available in the R package *lavaan* (Rosseel, 2012). A series of goodness-of-fit indices were computed and used to evaluate model fit: the chi-square to degrees of freedom ratio (χ^2/df), the root mean square error of approximation (RMSEA), the comparative fit index (CFI), the Tucker Lewis index (TLI), and the standardized root mean square residual (SRMR). Cut-off values for fit were considered good if χ^2/df was less than 2 (Schermelleh-Engel, Moosbrugger, & Muller, 2003), *CFI* and *TLI* > 0.95, *RMSEA* was less than 0.06, *SRMR* was less than 0.05, and *RMSE* was less than 0.08 (Hu & Bentler, 1999).

The most plausible model was selected based on goodness-of-fit, difference between CFIs of the two competitive models, and magnitude of correlations across factors as well as extant theory, and was subsequently tested on the father items.

In addition, internal consistency was evaluated for both mother and father items via ordinal Cronbach's alpha computed on the polychoric correlation matrix (Zumbo, Gadermann, & Zeisser, 2007) and McDonald's Omega computed on the CFA. Based on the selected model, we examined measurement invariance (for both mother and father items) across age groups using MG-CFA. First, we tested configural invariance by allowing all structural parameters to remain free; then, we simultaneously tested metric and scalar invariance by constraining factor loadings and thresholds to be equal across groups (Muthén & Muthén, 2010). If metric and scalar invariance are simultaneously reached, it implies that the meaning of the construct (the factor loadings) and the levels of the underlying items (thresholds) are equal in both groups, which therefore can be compared on their latent variable scores. The difference in CFI (Δ CFI) and RMSEA (Δ RMSEA) was computed between the two models (i.e., configural vs. metric and scalar). In addition, several fit indexes (i.e., CFI, TLI, RMSA) were inspected. To evaluate invariance between younger and older children, we considered the general guidelines proposed by Chen (2007), according to which a decrease of CFI less than 0.01 (Δ CFI), a RMSEA less than 0.015 (Δ RMSEA) between models, and acceptable model fit indices are considered evidence for model invariance (Cheung & Rensvold, 2002). All models were estimated using the *lavaan* package (Rosseel, 2012), and missing data (< 0.01) were handled with the pairwise method for WLSMV available in the R package lavaan (Rosseel, 2012).

Once scalar invariance was supported, potential differences on the anxiety and avoidance scores between younger and older children were evaluated. Because latent means are better indicators of potential differences than observed means (Brown, 2006), latent mean analysis using structural equation modeling was performed (separately for mother and father items). Specifically, we constrained loadings, thresholds, and error variances as well as the factor means of the younger group to be zero, whereas in the older group, means were allowed to vary. Because the latent means of the younger group were fixed to zero, the latent means of the older group represent the mean differences between the two groups (Finch & French, 2015). Additionally, following the procedure described by Finch and French (2015), to evaluate the effect sizes of the latent mean differences, we computed Cohen's *d*. According to Cohen (1988), a value of $d = 0.2$ is considered a small effect, $d = 0.5$ is considered a medium effect, and $d = 0.8$ is considered a large effect.

To test concurrent validity of the best fitting model, we performed a series of structural equation models (SEM) and evaluated the association of each latent attachment dimension (i.e., anxiety and avoidance) with the latent means of perception of felt security (as assessed via the SS). Using the same procedure, we tested convergent validity by evaluating the association of the brief ECR-RC with global self-worth (measured with the

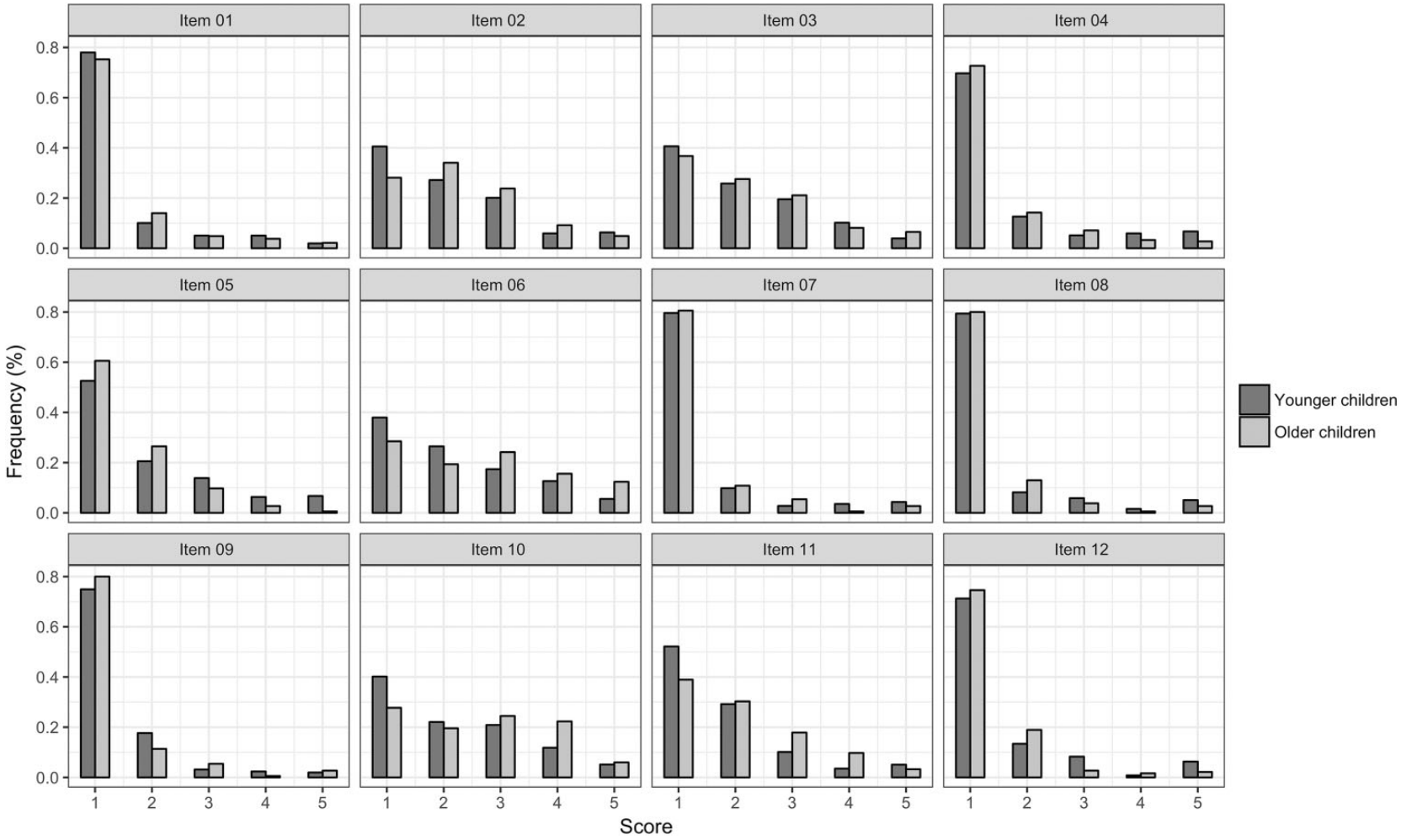

Figure 1. Response distributions of mother items.
Note. Younger children, $n = 259$; older children, $n = 185$.

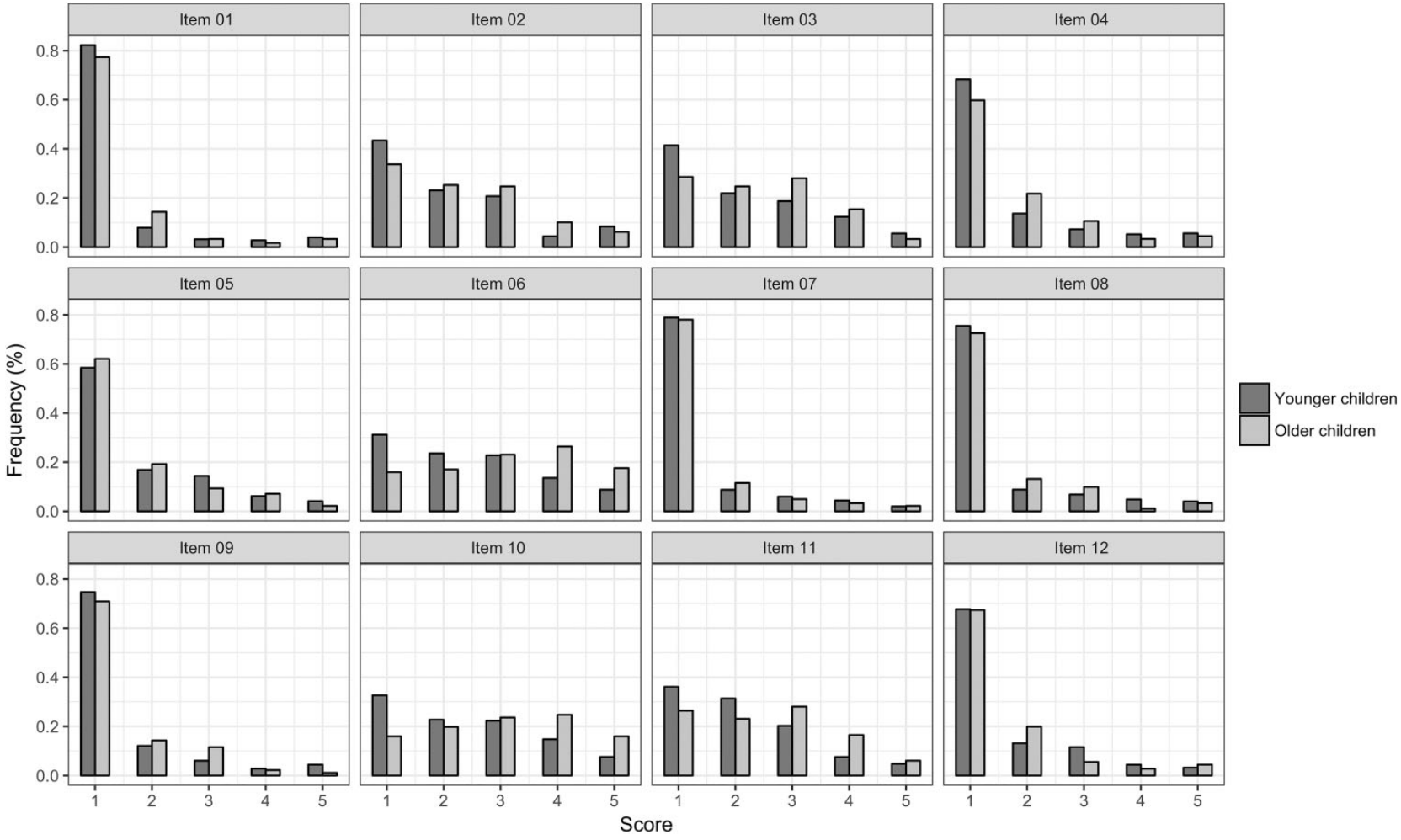

Figure 2. Response distributions of father items.
Note. Younger children, $n = 253$; older children, $n = 182$.

Table 1. Fit indices of the six confirmatory factorial models of the brief ECR-RC.

	$\chi 2$	*df*	*p*-value	*CFI*	*TLI*	*SRMR*	*RMSEA* [90% *CI*]
Mother							
Model 1: One-factor							
Younger children	289,787	54	< 0.001	0.850	0.817	0.152	0.130 [0.116–0.145]
Older children	310,824	54	< 0.001	0.938	0.924	0.189	0.161 [0.144–0.178]
Model 2: Two-factor							
Younger children	83,532	53	0.005	0.981	0.976	0.082	0.047 [0.026–0.066]
Older children	58,114	53	0.293	0.999	0.998	0.078	0.023 [0.000–0.053]
Model 3: Three-factor							
Younger children	63,836	51	< 0.107	0.992	0.989	0.058	0.031 [0.000–0.053]
Older children	55,534	51	< 0.308	0.999	0.999	0.077	0.022 [0.000–0.053]
Father							
Model 4: Two-factor							
Younger children	119.417	53	< .001	0.978	0.973	0.091	0.071 [0.054–0.087]
Older children	84.578	53	0.004	0.996	0.995	0.081	0.057 [0.033–0.080]

Note. Mother form: younger children, $n = 259$; older children, $n = 185$. Father form: younger children, $n = 253$; older children, $n = 182$.
df: degrees of freedom; CFI: comparative fit index; TLI: Tucker-Lewis index; SRMR: standardized root mean square residual; RMSEA: root mean square of approximation; WRMR: weighted root mean square residual.

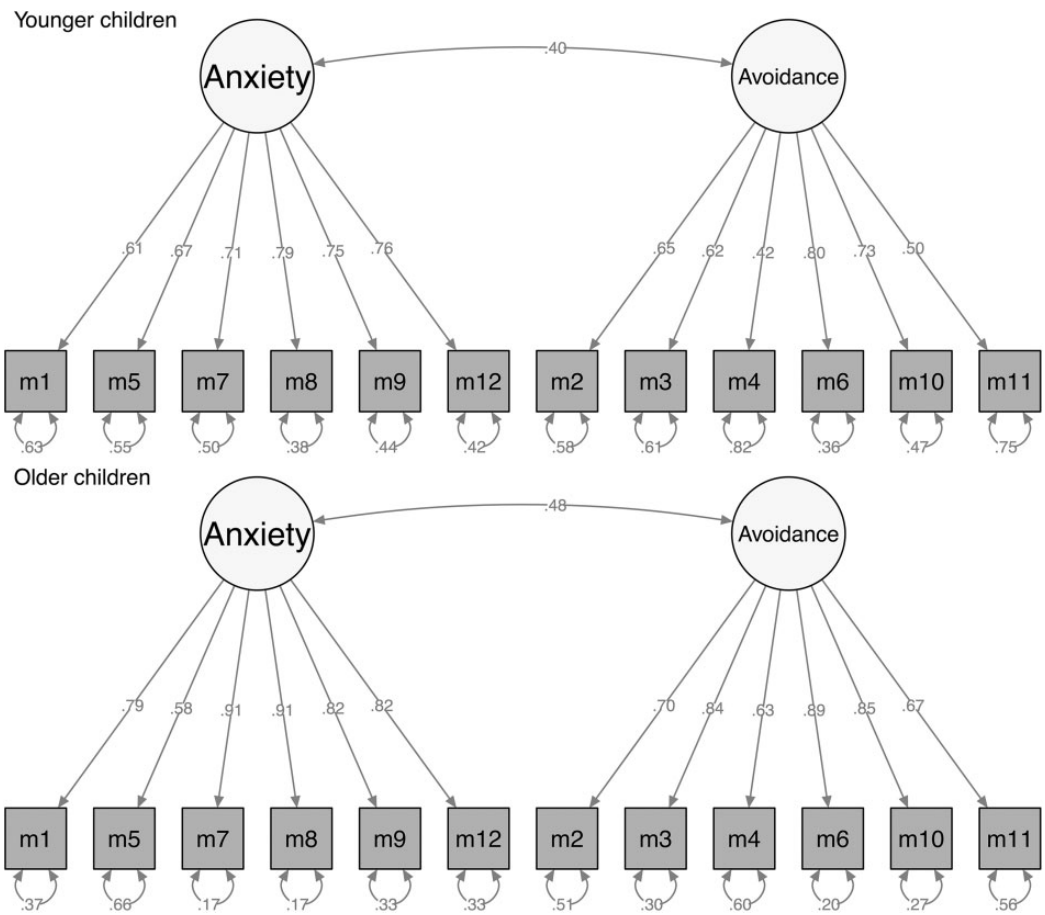

Figure 3. Best-fitting confirmatory factor model of the mother items.
Note. Younger children, $n = 259$; older children, $n = 185$. All structural coefficients are standardized. All factor loadings are significant at the 0.05 level.

SPPC) and the two ER strategies (i.e., CR and ES) assessed using the ERQ-CA. Furthermore, a series of SEMs (i.e., one for each outcome variable) were performed to evaluate the degree to which the ECR-RC subscales significantly predicted global self-worth and the two ER strategies after controlling for child gender and age.

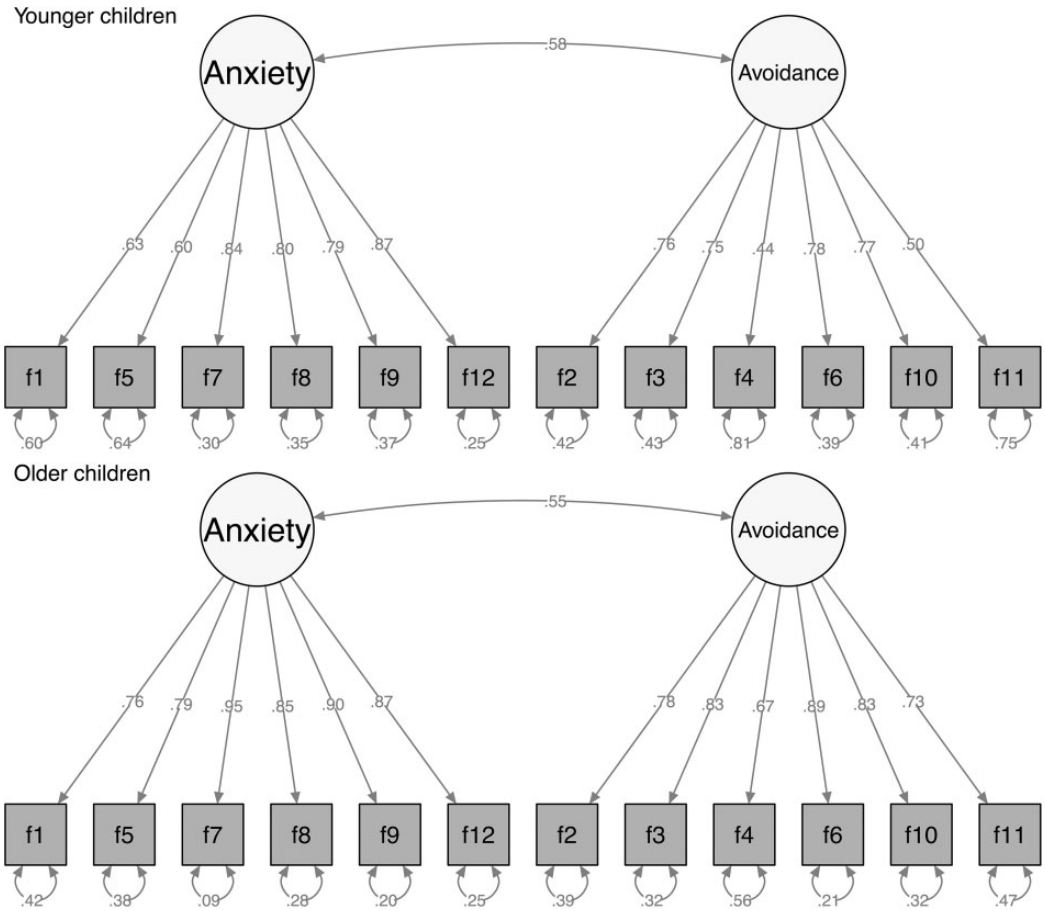

Figure 4. Best-fitting confirmatory factor model of the father items.
Note. Younger children, $n = 253$; older children, $n = 182$. All structural coefficients are standardized. All factor loadings are significant at the 0.05 level.

Results

Preliminary Analyses

CFAs of the nine items of the Italian SS used in this study yielded a one-factor solution, which showed a good fit to the data (mother: $\chi^2(27) = 32.851$, $p = 0.202$, $CFI = 0.997$, $RMSEA = 0.023(0.00–0.047)$; father: $\chi^2(27) = 28.512$, $p = 0.385$, $CFI = 1.00$, $RMSEA = 0.012(0.00–0.041)$).

Descriptive Statistics

Item response distributions of the brief ECR-RC by age group are presented in Figure 1 (mother) and Figure 2 (father). In both cases, most of the items showed a right-skewed distribution, further supporting the use of the WLSMV estimator (Rhemtulla et al., 2012). More detailed descriptive statistics for each item are reported in Tables A1 and A2 of the Appendix section.

Factor Structure and Invariance Across Middle-Childhood and Early Adolescence

Initial CFA results for the one-factor model tested on the mother items (i.e., Model 1) separately for younger and older children yielded a poor fit in both samples (see Table 1). After testing for the two-factor solution (Model 2, Table 1), the model yielded a good fit in all the considered fit indices both in the younger and the older samples. Likewise, the three-factor solution (Model 3, Table 1) fitted the data well in both groups of children.

Comparison of the two- and three-factor solutions of the mother items suggested that in the three-factor model, there was only a slight increase of fit indices for younger children (Δ CFI = 0.011), and equivalent fit indices for older children (Δ CFI = 0). In the two-factor solution, correlations between the anxiety and avoidance subscale scores were $r = 0.40$ and $r = 0.48$ for younger and older children, respectively (Figure 3). Noteworthy, extraction of the security factor (three-factor model) did not reduce the correlation between anxiety and avoidance dimensions (younger children: $r = 0.41$; older children $r = 0.50$). Furthermore, in the three-factor solution, correlations between avoidance and the security factor were very high ($r = -0.72$ and $r = -0.93$ in younger and older children, respectively), making it difficult to establish the existence of two separate factors.

Therefore, based on extant theory, we considered the two-factor model more adequate in our sample. The two-factor model tested on the father items (i.e., M4) also yielded a good fit in both younger and older children (see Table 1). Similar to the mother model, high correlations between the avoidance and anxiety factors were found (see Figure 4). Table A3 in Appendix I provides factor loadings of the two-factor model for mother and father items. Anxiety scores in

Table 2. Model fit for the two-factor model tested for invariance across middle childhood and early adolescence.

	$\chi 2$	*df*	*p*-value	*CFI*	*TLI*	*SRMR*	*RMSEA* [90% *CI*]	Δ *CFI*	Δ *RMSEA*
Mother form									
STEP 1 – Configural	141,645	106	0.012	0.994	0.992	0.081	0.039 [0.019–0.055]		—
STEP 2 – Metric and scalar	212,214	152	< 0.001	0.989	0.991	0.087	0.042 [0.028–0.055]	0.004	0.003
Father form									
STEP 1 – Configural	203,995	106	< 0.001	0.991	0.989	0.087	0.065 [0.052–0.079]		—
STEP 2 – Metric and scalar	263,882	152	< 0.001	0.990	0.991	0.089	0.058 [0.046–0.070]	0.001	0.007

Note. Mother form: younger children, $n = 259$; older children, $n = 185$. Father form: younger children, $n = 253$; older children, $n = 182$.
Δ CFI: difference among CFIs; Δ RMSEA: difference among RMSEAs.
df: degrees of freedom; CFI: comparative fit index; TLI: Tucker-Lewis index; SRMR: standardized root mean square residual; RMSEA: root mean square of approximation; WRMR: weighted root mean square residual.

Table 3. Means and SD and Polychoric Alphas and Omegas for the two ECR-RC subscales, Anxiety and Avoidance, by age group.

	Younger children						Older children					
	n	Range	Mean	SD	Alpha *[95% CI]*	Omega *[95% CI]*	*n*	Range	Mean	SD	Alpha *[95% CI]*	Omega *[95% CI]*
(1) Anxiety (Mother)	259	1–5	1.53	0.70	0.87 [0.84–0.89]	0.81 [0.75–0.88]	185	1–5	1.40	0.63	0.91 [0.89–0.93]	0.72 [0.65–0.79]
(2) Avoidance (Mother)	259	1–5	2.02	0.75	0.77 [0.72–0.81]	0.91 [0.84–0.95]	185	1–5	2.22	0.90	0.88 [0.85–0.91]	0.82 [0.75–0.88]
(3) Anxiety (Father)	253	1–5	1.52	0.73	0.89 [0.87–0.91]	0.86 [0.82−0.90]	182	1–5	1.51	0.77	0.93 [0.91–0.95]	0.81 [0.75–0.89]
(4) Avoidance (Father)	253	1–5	2.16	0.83	0.79 [0.75–0.83]	0.92 [0.88–0.96]	182	1–5	2.50	0.93	0.87 [0.84–0.90]	0.91 [0.85–0.98]

Note. ECR-RC: Experience in Close Relationships Scale-Revised Child version subscale; SD: standard deviation; CI: confidence interval.

relation to mother and father correlated positively, as did avoidance scores (see Table 4).

In order to test invariance across the two age groups, MG-CFA on the two-factor solution was performed following two steps. In step 1, configural invariance (without parameter restrictions) of the two-factor model revealed a good model fit to the data (Table 2), suggesting that the factor structure was similar between younger and older children for both parents. Because configural invariance was supported, in the subsequent step we held loadings and thresholds invariant across groups. The fit of these models was also good (see Step 2, Table 2). The chi-square difference was not significant, and both Δ CFI and Δ RMSEA values between the constrained and unconstrained models were less than 0.01 (see Table 2), showing that metric and scalar invariance were supported. Table 2 reports the results of model fits and comparisons.

In both samples, the subscales of the two-factor model showed good internal reliability, with polychoric Cronbach's alphas ranging from 0.77 to 0.93 and McDonald's Omegas ranging from 0.72 to 0.92, with slightly better estimates for the father items (see Table 3).

Because scalar invariance was reached, two latent mean analyses were conducted (i.e., mother and father items separately) to estimate potential differences in anxiety and avoidance scores between the two age groups. Younger children were chosen as a reference group constrained to have a mean of zero, whereas the older children's mean was allowed to vary. Results indicated that older children reported significantly higher latent mean values in attachment avoidance to mother than their younger counterparts, although no difference in anxiety was found (see Table 4). Similarly, with regard to father items, older children reported significantly higher latent mean values in avoidance compared to younger children. The structured mean differences between groups in anxiety and avoidance scores are presented in Table 4. Cohen's *d* indices indicate that the values of effect size ranged from −0.06 to 0.52 (Table 4).

Table 4. Structured mean differences between age groups on the brief ECR-RC subscale scores.

	Younger children[a]	Older children	*p-value*	Effect size
Latent variable	Latent mean	Latent mean		
Anxiety (mother)	0	−0.039	0.747	−0.056
Avoidance (mother)	0	0.259	< 0.001	0.423
Anxiety (father)	0	0.048	0.625	0.073
Avoidance (father)	0	0.372	< 0.001	0.518

Note. Mother items: younger children, $n = 259$; older children, $n = 185$. Father items: younger children, $n = 253$; older children, $n = 182$.
ECR-RC: Experience in Close Relationships Scale-Revised Child version subscale; SD: standard deviation; CI: confidence interval.
[a]Reference group.

In terms of concurrent validity, in each of our SEMs (i.e., younger and older children), we found significant negative correlations between perceived security scores obtained via the SS and the anxiety and avoidance subscales of the brief ECR-RC (see Table 5).

With regard to convergent validity, as shown in Table 5, anxious and avoidant attachment scores toward mother and father were negatively related to self-worth in both younger and older children (all $ps < 0.001$). Avoidance toward mother and father was positively associated with ES in both younger and older children; anxiety toward parents showed a similar pattern, but only among younger children. Moreover, avoidance toward both mother and father was statistically and negatively related to the use of CR in younger, but not in older children. Correlations among study variables (standardized parameter estimates on latent scores) are summarized in Table 5.

Table 5. Latent score correlations between study variables.

	Younger children						Older children					
	Mother			Father			Mother			Father		
	n	Anxiety	Avoidance	*n*	Anxiety	Avoidance	*n*	Anxiety	Avoidance	*n*	Anxiety	Avoidance
Convergent validity												
Felt security to mother	253	−0.522***	−0.749***				185	−0.682***	−0.880***			
Felt security to father				253	−0.389***	−0.814***				181	−0.646***	−0.826***
Concurrent validity												
General self-worth	250	−0.506***	−0.428***	243	−0.369***	−0.502***	185	−0.515***	−0.649***	181	−0.407***	−0.472***
Expressive suppression	229	0.370***	0.572***	224	0.337**	0.459***	185	0.138	0.647***	182	0.179	0.394***
Cognitive reappraisal	229	−0.025	−0.350***	222	−0.082	−0.220**	185	0.032	−0.116	182	−0.019	−0.117

Note. **$p < 0.05$; ***$p < 0.001$.

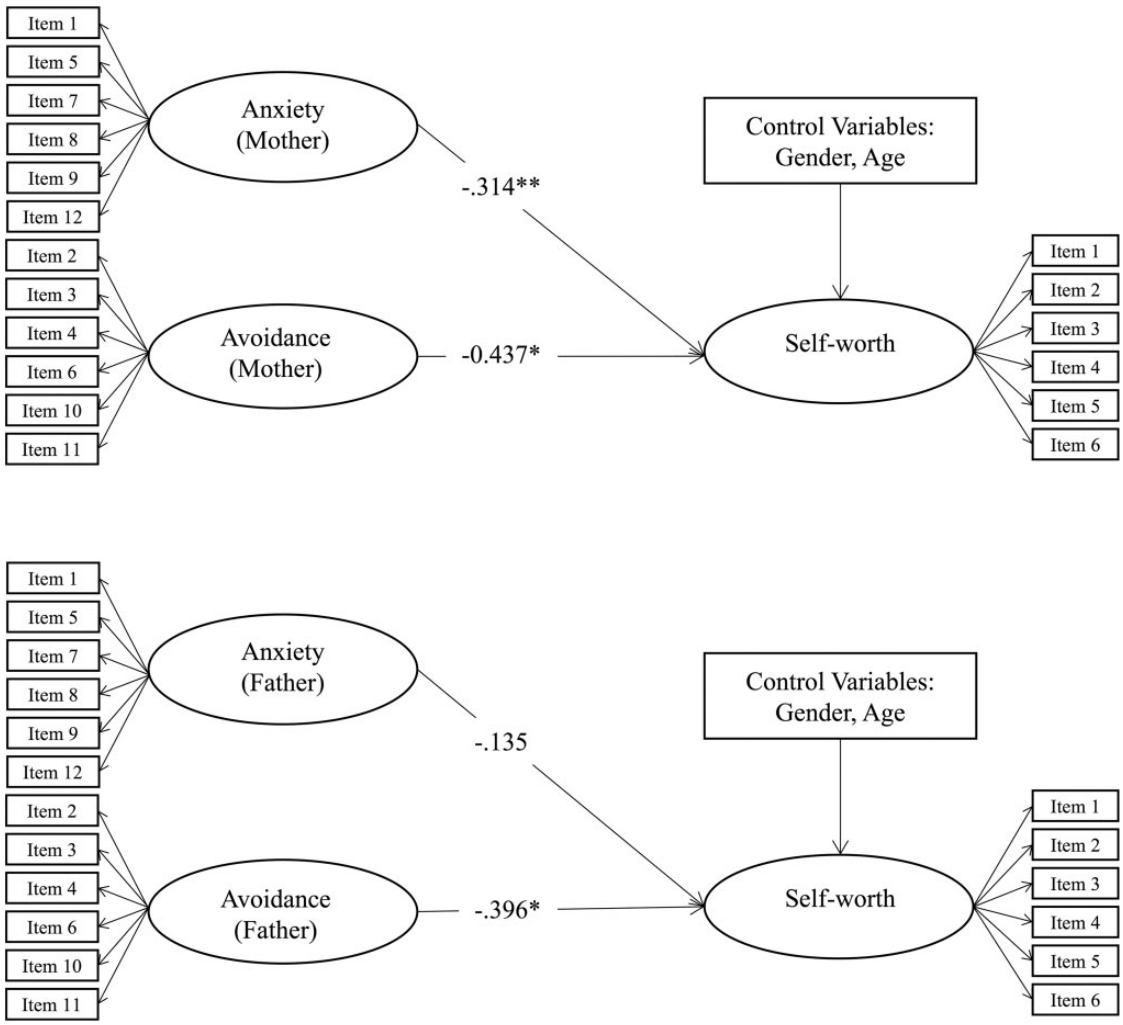

Figure 5. Structural equation model of the association between attachment dimensions (anxiety and avoidance, mother and father items) and self-worth.
Note. Mother, $n = 435$; father, $n = 424$. All structural coefficients are standardized.
*$p < 0.05$; **$p < 0.01$.

Lastly, three separated SEMs for each parental figure (i.e., mother and father) were performed on the entire sample, with attachment-related dimensions as independent variables, each of the criterion variables (i.e., self-worth, ES, CR) as dependent variables, and gender and age as control variables. Anxiety and avoidance toward mother, and avoidance toward father, were significantly and negatively related to global self-worth ($\beta = -0.314$, $p = 0.009$; $\beta = -0.437$, $p = 0.022$, and $\beta = -0.396$, $p = 0.011$, respectively; see Figure 5).

In relation to ER strategies, avoidance scores toward mother and father were positively associated with the use of ES

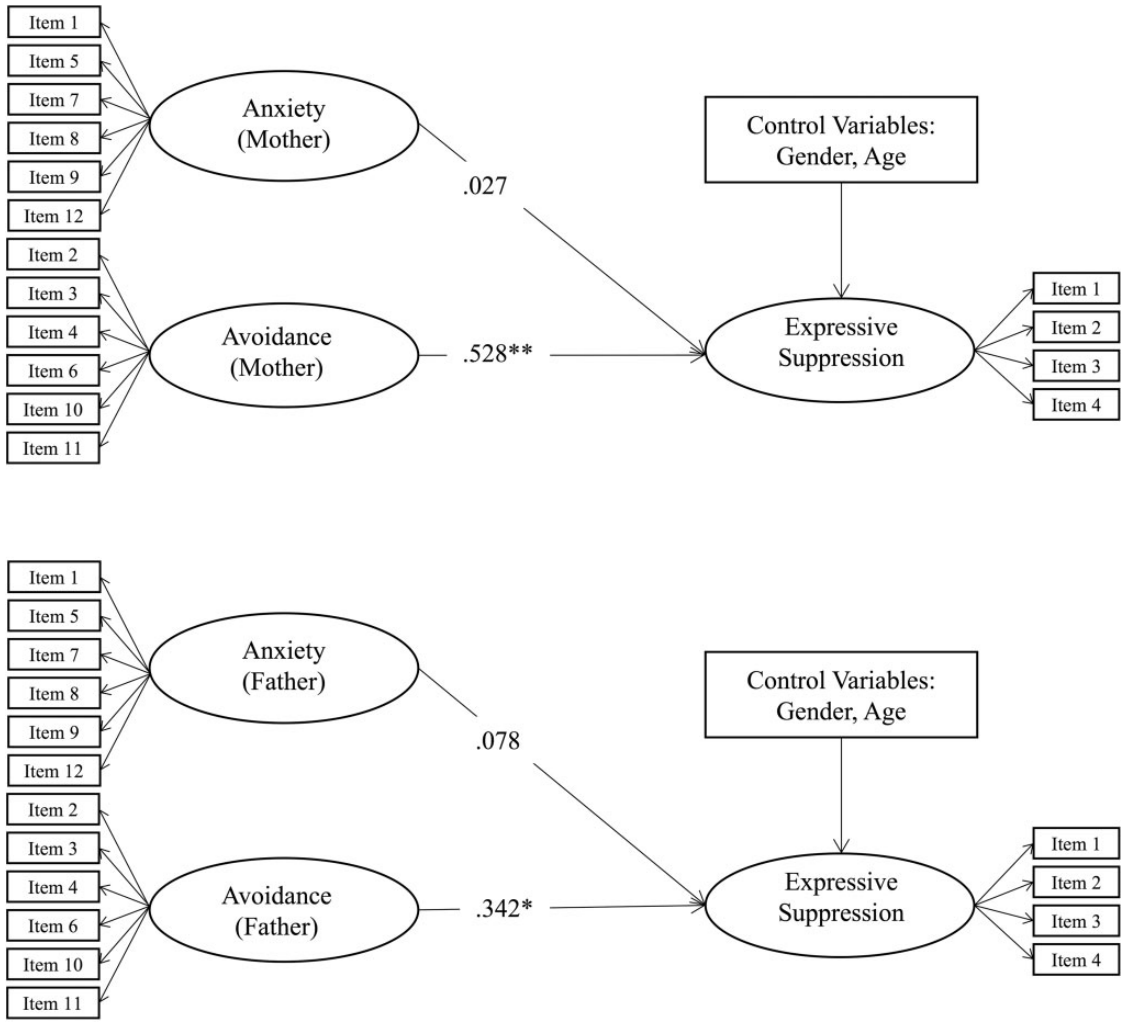

Figure 6. Structural equation model of the association between attachment dimensions (anxiety and avoidance, mother and father items) and expressive suppression.
Note. Mother, $n = 414$; father, $n = 406$. All structural coefficients are standardized.
$^{*}p < 0.05$; $^{**}p < 0.01$.

($\beta = 0.528$, $p = 0.003$, and $\beta = -0.342$, $p = 0.047$, respectively). All patterns of associations are depicted in Figures 6 and 7.

Discussion

The study of attachment in middle childhood is a relatively new research field. Although many theoretical advances have been made concerning the main features of attachment in this developmental period, different opinions still exist on how to measure it appropriately. This study aimed to test the psychometric properties of a short form of the ECR-RC, a recently developed questionnaire designed to assess anxiety and avoidance dimensions by examining the child's relationship with his/her mother and father. To make the scale more understandable for younger children, we reduced the response options from 7 to 5 and labeled all points of the scale.

In terms of factor structure, CFAs were conducted for age subgroups (8–10 years vs. 11–13 years) to examine fit of the two-factor model proposed by Brenning et al. (2014) and the three-factor model proposed by Lionetti et al. (2017). Overall, analyses revealed that both models yielded acceptable fit indices. Yet, comparison of the two- and three factor solutions suggested that there was only a slight increase in fit for the three-factor model (and only in the younger age group), thus lending support to the plausibility of the original two-factor structure of the ECR-RC (i.e., attachment anxiety and attachment avoidance). In both mother and father items, the scales for anxiety and avoidance showed good internal consistency. Also, consistent with prior work testing the factor structure of the ECR-RC (see Brenning et al., 2014; Skoczeń, Głogowska, Kamza, & Włodarczyk, 2018) as well as with the results of studies using the ECR and the ECR-R with adults (see Cameron, Finnegan, Morry, 2012 for an overview), a positive correlation between the two factors emerged in both age groups. Although this result is in contrast with the view of anxiety and avoidance as orthogonal (see Mikulincer, Shaver, Sapir-Lavid, & Avihou-Kanza, 2009), the latter assumption gained less empirical evidence in research even with the ECR (Cameron, Finnegan, & Morry, 2012). As argued by Fraley, Heffernan, Vicary, & Brumbaugh (2011), a conceptual distinction between constructs does not imply a complete statistical

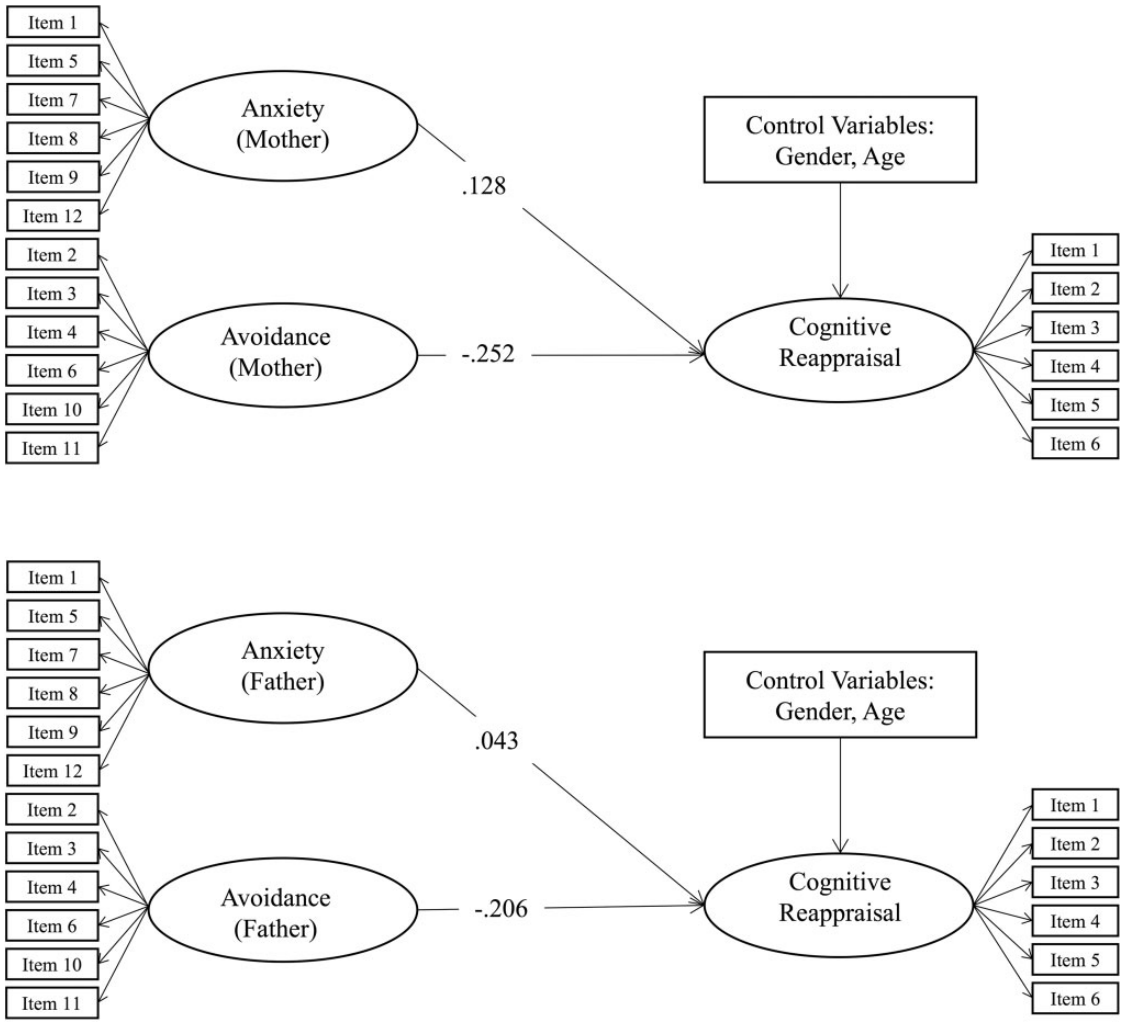

Figure 7. Structural equation model of the association between attachment dimensions (anxiety and avoidance, mother and father items) and cognitive reappraisal.
Note. Mother, $n = 414$; father, $n = 404$. All structural coefficients are standardized.

independence between them. Hence, our result might reflect an intercorrelation between avoidance and anxiety dimensions within the attachment regulation system, in which anxiety could be considered a monitoring system and avoidance a behavioral orientation system (Fraley & Shaver, 2000). In this perspective, for example, avoidance of intimacy could be a consequence of anxiety about rejection, and thus the mutual feedback between attachment-related dimensions may translate into a positive intercorrelation (Conradi, Gerlsma, van Duijn, & de Jonge, 2006).

With regard to invariance of the brief ECR-RC across middle childhood and early adolescent samples, MG-CFAs indicated that the questionnaire measured attachment anxiety and avoidance in the same way for younger and older children at the item and at the scale level, thus corroborating the findings reported by Brenning et al. (2011) for the long form of the ECR-RC.

Examination of differences in anxiety and avoidance latent scores for father and mother items revealed lower levels of anxiety and higher levels of avoidance from middle childhood to early adolescence. This pattern may reflect the normative process of increased adolescent autonomy from parental figures (Ammaniti, van Ijzendoorn, Speranza, & Tambelli, 2000).

Finally, we examined concurrent and convergent validity by correlating scores for each of the brief ECR-RC dimensions with perceived security as assessed via the SS, global self-worth measured via the relevant sub-scale included in the SPPC, and ER strategies (i.e., ES and CR) assessed by the ERQ-CA. As expected, lower scores in both anxiety and avoidance toward mother and father were associated with higher scores in perceived security. Also, consistent with previous studies (see Brenning et al., 2015), higher scores in the self-worth scale were associated with lower levels of anxiety and avoidance toward mother and father. This finding is in line with attachment theory, which posits that attachment has important implications for the development of the self: secure children construct a positive working model of themselves and others, and thus evaluate themselves more favorably than insecurely attached children. Of note, the link between perceived parent attachment and child self-worth was slightly higher for younger than for older children. This result emphasizes the protective role of attachment in this developmental stage, which is characterized by many physical, cognitive, and social changes associated with puberty and often relates to a certain decline in self-esteem (Robins & Trzesniewski, 2005).

In relation to ER strategies, it is well-known that the attachment bond represents the main context in which children understand and learn from their emotional experiences (Cassidy, 1994). As expected, we found a significant and positive association between avoidant attachment and the suppression of emotional expression, also after controlling for child age and gender. This finding is consistent with the theoretical view that children with an experience of repeated rejection are more likely to develop an insecure, avoidant attachment style and learn to manage their emotions so as to reduce future rejection by decreasing expressive behavior (Cassidy, 1994). Future research may address how this association develops over time and is modeled by other features of children's social ecology.

With regard to CR, no significant associations emerged with the two attachment-related dimensions. Yet, prior research reported negative correlations between the anxiety and avoidance subscales of the ECR and CR among adults (i.e., Poncy, 2017). Similarly, Gresham and Gullone (2012) found a bivariate association between perceptions of relationships with parents based on the three dimensions of the Inventory of Parent and Peer Attachment (Armsden & Greenberg, 1987) and the endorsement of CR in children and adolescents. A possible explanation for our findings is that the use of this strategy—which is considered to be highly adaptive—increases with age and experience (John & Gross, 2004). Albeit longitudinal studies are warranted to establish causality, it may be hypothesized that the link between perceived insecure attachment (in terms of anxiety and/or avoidance) and the tendency to cognitively reframe potentially emotion-eliciting events becomes more evident starting from late adolescence due to enhanced cognitive complexity and maturation. Moreover, it should be noted that the use of attachment measures assessing different dimensions (i.e., perceived security with the IPPA, and anxiety and avoidance with the ECR) may result in diverse patterns of associations with ER strategies. Thus, our data provide support—at least in part—for convergent validity of the brief ECR-RC with the ERQ-CA, but more studies are needed to shed light on the hypothesized link between anxious/avoidant attachment and the CR subscale at different ages.

To summarize, the Italian short form of the ECR-RC appears to be psychometrically sound and shows the expected relationships with variables known to be associated with attachment in middle childhood. Importantly, our analyses indicate that it is an appropriate tool to assess attachment anxiety and avoidance also in younger children (8-10 yrs). In this regard, the change in response format (from 7 to 5 points) might have reduced the potential bias originating from a different interpretation of the scale, and the labeling of each response option may have facilitated children's comprehension (Borgers et al., 2003). However, it should be noted that in this study, the ECR-RC was not directly compared with the seven-point Likert version previously used in other studies. Such comparison may provide some guidance as to which scale to use in relation to age. A likely outcome of such work might be that the ECR-RC based on five points is more appropriate for younger children, whereas the seven-point scale measure is more appropriate in late adolescence. Further studies are needed to clarify this issue.

Although our study provided further evidence for the validity of the brief ECR-RC in its use with 8- to 13-year-old children, we acknowledge some limitations. First, replication studies in other cultures are needed to allow greater confidence in the generalizability of results. Second, examination of test-retest reliability of the brief ECR-RC scale scores in future research will prove valuable in determining the stability of the scale. Third, this study is limited by the exclusive reliance on self-report measures to test concurrent and convergent validity. Hence, future research should include other assessment methods (e.g., automatic process measures) and sources of information (e.g., parents, teachers) to assess concurrent, convergent, and predictive validity of the ECR-RC. Also, we used only nine of the 12 items included in the Italian version of the SS to test concurrent validity. Further research should replicate this finding by using the full pool of items. Fourth, this study assessed ER strategies based on a specific theoretical framework (i.e., process-oriented model; Gross, 1998). Following previous work (see Brenning et al., 2011), it may be useful to include a measure of hyperactivating ER strategies (e.g., dysregulated expression) to obtain a more comprehensive picture of convergent validity between anxious attachment and the endorsement of specific ER strategies. Lastly, because we recruited our sample in schools, participants represented a "normative" sample. Future studies should administer the ECR-RC to at-risk groups to test its discriminant validity in middle childhood,

Despite these shortcomings, these findings provide evidence that the short form of the ECR-RC is an age-appropriate measure to assess anxiety and avoidance in Italian children aged between 8 and 13 years. Specifically, the ECR-RC is likely to provide utility in further research interested in examining the features and role of attachment in middle childhood and early adolescence.

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Supplemental material

The supplemental material is available online with the article.

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